



Improving Power-Inductor Performance by Mixing Sub-micro Fe Powder with Amorphous Soft Magnetic Composites

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Amorphous soft magnetic powder is one of the most favorable materials for power applications working in the MHz range. In order to induce high permeability, DC bias characteristics, and low core loss in the working frequency range, the core must achieve high packing density. Sub-micro-sized α -Fe powder has been applied to inductors because of its high saturation magnetization and low intra-particle eddy-current loss. In this study, we investigated the magnetic properties of amorphous Fe alloy/sub-micro-sized α -Fe powder composites. We prepared composites as a bimodal toroid core with various mixing ratios of amorphous Fe alloy and sub-micro α -Fe, whose average particle sizes are approximately 35 μm and 3.3 μm , respectively. As the amounts of sub-micro α -Fe contents increase, the packing ratio and density of the core maximize at the cores, which have a sub-micro α -Fe content of 30 wt.%. The measurement results of the magnetic relative permeability (μ) and quality (Q) at 1 MHz revealed similar behaviors with the packing ratio and density; both μ and Q are a maximum of 35.5 and 47 in the core with 30 wt.% sub-micro α -Fe contents. The total evaluated core-loss (P_{cv}) under a driven magnetic flux 200 mT in the MHz range gradually decreased as the sub-micro α -Fe content increased up to 30 wt.%, and the minimum core-loss was 562.2 mW/cm³ at 1 MHz. Based on the experimental results, the improved magnetic performance of the core can be obtained not only by the intrinsic properties of material but also by the microstructure of the core body.

Key words: Soft magnetic composite, Fe based amorphous alloy, Power application, Ollendorf's equation, Bimodal core

INTRODUCTION

Recently, soft magnetic materials have been applied in magnetic devices and power applications such as chip power inductors or transformers. Soft magnetic materials with high saturation magnetization (M_s), high permeability, and low core loss are essential for induction of high power with superior energy efficiency and also for a miniaturization.^{1,2} Therefore, spinel ferrites with high permeability and crystalline metal alloys with high saturation

magnetic flux density have been applied widely in the past. But each soft magnetic material has intrinsic drawbacks. Spinel ferrite cannot be applied to a high-frequency electrical device since it has a low magnetic resonance frequency. In the case of crystalline metal alloy, the energy efficiency drops sharply with an increase in working frequency due to increased core loss induced by low electrical resistivity.³ These limitations have triggered a demand for new materials applicable to electrical devices (such as mobile devices) working in the MHz range. Amorphous Fe based alloys have been found to overcome these challenges. Physically, the amorphous phase has a higher electrical resistivity (several hundred $\mu\Omega\cdot\text{cm}$) than a

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crystalline alloy. It is the most important physical parameter influencing the eddy current loss.⁴ However, despite these advantages, the suppression of magnetic flux density is a challenge to overcome because of the additives used to enhance amorphous forming capabilities.

At present, soft magnetic composites of mixed soft magnetic materials with insulating polymer are interesting alternative materials because of their viable overall permeability and remarkable contributions to prevention of inter-particle eddy-current loss (P_e).^{3,5,6} Especially, a bimodal composite of magnetic powders with different sizes is a viable candidate to increase the high permeability and magnetization resulting from high packing density.⁷⁻⁹

To the best of our knowledge, there are no reports on increasing permeability and decreasing loss successfully by controlling the microstructure of cores. In this study, we selected amorphous Fe alloys a few tens of microns in size as the major powder and sub-micro-size Fe with high magnetization as the minor powder. We investigated the micro structure, AC magnetic properties, and core loss of $(100 - x)$ wt.% amorphous Fe alloy/sub-micro Fe composite ($x = 0-40$).

EXPERIMENTAL SETUP

Amorphous Fe alloy (A-Fe alloy) and sub-micro-sized α -Fe powder (S-Fe) were prepared by gas atomizing. A-Fe alloy powders were distributed by sieves. In order to exclude powder size effect for electromagnetic properties, we selected A-Fe alloy powders distributed in 25–53 μm by sieving. A-Fe alloy and S-Fe powder were mixed in weight ratios of 100:0 to 60:40. The powder mixture was blended with a 2.5 wt.% binder containing insulating epoxy, dispersant, and hardening agent in solvent. The blended mixture was thoroughly mixed and then dried in an agate mortar. The dried mixtures were

granulated manually. To make a green body core, a toroidal-shaped die was used. The applied pressure is fixed to 1.5 metric tons per cubic centimeter to exclude the effect of pressure on the microstructure. The green bodies were cured at 160°C for 2 h under ambient air.

The particle size of the prepared powder was analyzed by particle size distribution (PSD). Crystal phase and microstructure were analyzed by x-ray diffraction (XRD) using Cu $k\alpha$ ($\lambda = 1.54046 \text{ \AA}$) and scanning electron microscopy (SEM). The DC magnetic properties were measured by a vibration sample magnetometer (VSM). Frequency dependency of relative permeability and quality were investigated by using impedance analyzer measurement. Core loss was measured by a $B-H$ analyzer at various frequencies and applied fields. The fabricated toroidal cores were wound 10 times by using Cu wire for impedance analyzer and $B-H$ analyzer measurements.

RESULTS AND DISCUSSION

Figure 1 shows the particle size distribution and morphology of the prepared powders. Figure 1a shows that both prepared powder samples have different size distributions. A-Fe alloys had particle sizes of D10, 17.9 μm ; D50, 34.7 μm ; and D90, 47.7 μm . S-Fe powder distribution was measured as D10, 1.73 μm ; D50, 3.55 μm ; and D90, 7.12 μm . The average particle sizes of the A-Fe alloy and S-Fe were approximately 35 μm and 3.5 μm , respectively. The SEM image (Fig. 1b) shows that the particles of both prepared powders were uniformly spherical and confirmed the results of PSD measurements. Based on the particle packing theory, a cubic body having one sized spherical particle induces maximum density when its particles are perfectly layered in the body-centered cubic with a maximum packing density of about $\sim 74\%$.¹⁰ Suzuki et al.¹¹

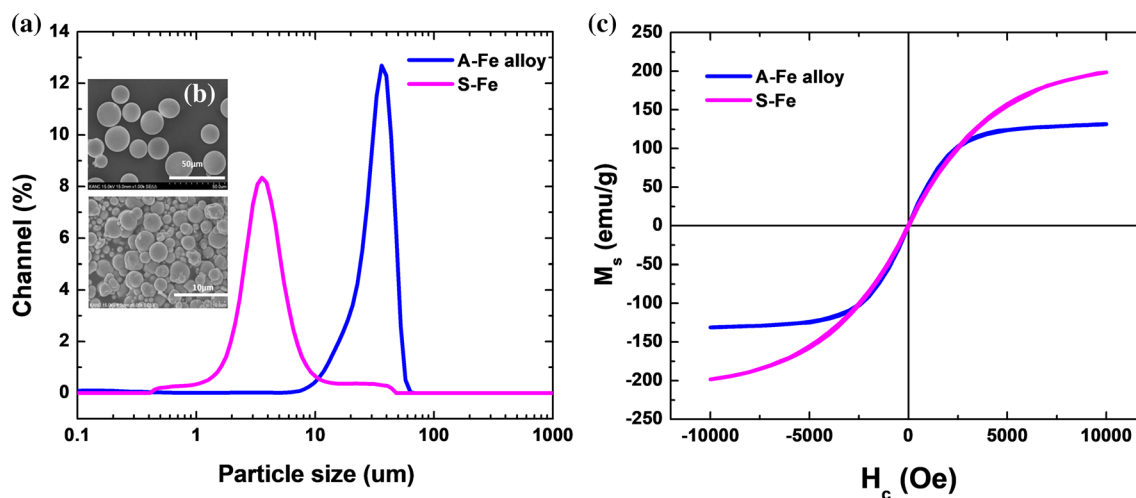


Fig. 1. The particle size distribution (a), morphology (b), and magnetic hysteresis curves (c) of the prepared magnetic A-Fe alloy and S-Fe powders.

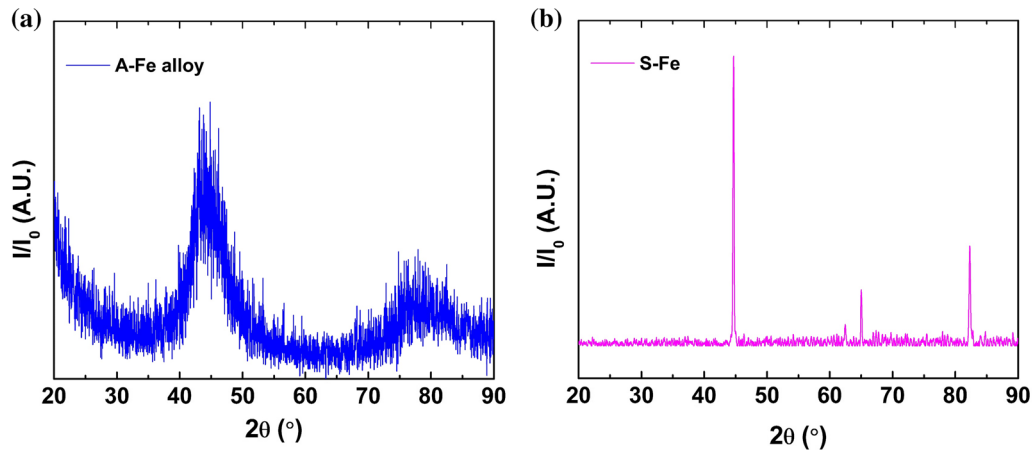


Fig. 2. The x-ray diffraction patterns of A-Fe alloy (a), and S-Fe powder (b).

suggested that the packing density of bulk body accumulated by particles be defined as the total volume excluding the porosity. Based on previous studies, addition of S-Fe powders can reduce the porosity of cores through distribution within A-Fe alloy particles, which results in improved core packing density. From the energy dispersive x-ray spectroscopy (EDS) measurement, we can assume that the composition of prepared A-Fe alloy is $\text{Fe}_{86.5}\text{Si}_{4.2}\text{Nb}_{3.4}\text{B}_{1.7}\text{P}_{1.5}\text{Mo}_{1.4}\text{C}_{1.3}$ (wt.%) (data not presented). Figure 1c shows the magnetic field-dependent magnetization of both prepared powders. The evaluated saturation magnetization and coercivity of S-Fe powder are 198 emu/g and 16 Oe, respectively. The saturation magnetization of the A-Fe alloy, 131 emu/g, is lower than that of S-Fe powder because of the small amount of Fe contents. The coercivity of A-Fe alloy reduces to 5 Oe. This is one of the main properties of amorphous metal alloys, in which the dislocation and grain boundary do not exist.^{3,12,13}

X-ray diffraction patterns are shown in Fig. 2. The pattern of the A-Fe alloy (Fig. 2a) shows that the first-order peak is around 43.2° . The broad pattern and weak peak intensity are typical characteristics of amorphous material. On this pattern, no impurity phase is observed except for the α -Fe phase. Conversely, the x-ray pattern of S-Fe (Fig. 2b) shows narrow peaks with high intensity. Based on the XRD pattern, we can observe that S-Fe powder has a crystalline structure.

Figure 3 shows the density and packing density of prepared bimodal cores according to mixing ratio. It indicates that the density and packing density of cores are totally dependent on the mixing ratio of the powders. In the core without S-Fe powder, there is the lowest density, of 5.35 g/cm^3 , and as the S-Fe content increases, the density of the bimodal core increases. The maximum core density, observed at the mixing ratio of 7:3, was 6.35 g/cm^3 , which is 19% higher than that of the core without S-Fe powder. Differences of core packing density increased

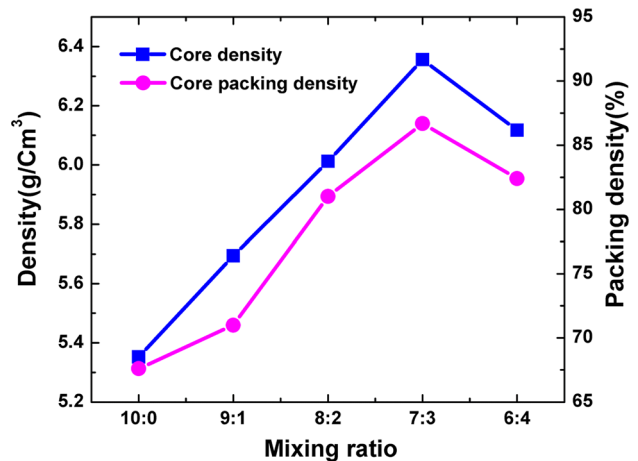


Fig. 3. The density and packing density of toroidal cores according to the mixing ratio.

similar to core density, and the maximum packing density was 86.7% at 30 wt.% S-Fe mixing ratios. In contrast, the density and packing density of a bimodal core containing 40 wt.% S-Fe powder decreased to 6.11 g/cm^3 and 82.4%, respectively. These behaviors resulted because the S-Fe powder, at a few microns in size, was packed between A-Fe powders with a few tens of microns in size.

Permeability is the most crucial characteristic of a power inductor, because the high μ of a core's material can improve the energy efficiency of the power application, such as a DC-DC converter. Figure 4 shows the frequency dependency of from 1 kHz to 110 MHz. All cores show stable permeability up to a frequency of approximately 20 MHz. Cores with a mixing ratio of 10:0 and 9:1 show a higher resonance frequency than those with other ratios, because of low contents of S-Fe powder. As S-Fe content increases, permeability of the bimodal cores increases. Inversely, the magnetic resonant frequency moves to under 100 MHz.

Table I. The electrical properties of cores in this study and reference

Sample	$\mu_{\max}$ @ $f \leq 1$ MHz	Core loss (mW/cm ³) @ 200 mT				References
		100 kHz	300 kHz	500 kHz	1 MHz	
70%A-Fe + 30%S-Fe	35.5	23.5	123.7	189.3	562.2	This work
FeCrSiB amorphous	~ 16	—	—	—	—	Ref. 5
CIP	~ 30	—	—	—	—	Ref. 18
85 wt.%FeSiB + 15 wt.%FeNi50	~ 35	—	—	—	—	Ref. 19
Megaflux	26–90	50–60	270–300	—	—	Ref. 20

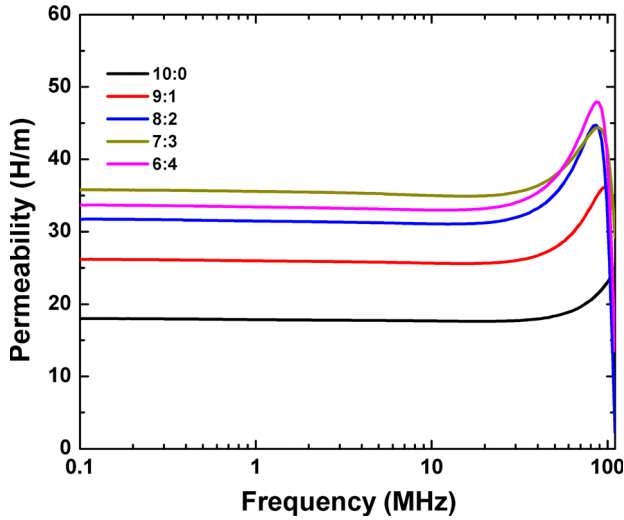


Fig. 4. The permeability as a function of applied frequency from 1 kHz to 110 MHz.

In Fig. 5, we represent the change of μ and Q of the prepared cores at a 1 MHz frequency. In the bimodal-core mixing ratio of 10:0 (without S-Fe powder), the μ is lowest, to 18, at 1 MHz. As the amounts of S-Fe increase, the core's μ also increases. The highest μ is obtained at 7:3, with the highest packing density, where the μ is 35.5; μ decreases for bimodal cores containing more of S-Fe powder than 30 wt.%. On the powder core, the packing density largely influences the μ of the powder core. Ollendorff's Eq. 1 shows the relation of packing density to the core's permeability.¹⁴ Based on this equation, the core's μ composed by magnetic powder is increased by increasing the packing density.^{14,15} Figure 5b shows the relationship between packing density and permeability of the prepared core together with calculated values. For this calculations, coefficient of diamagnetic field was approximately $(1/3)\exp(-3\eta)$.¹⁶ The above result is a clear indication that the improved permeability is as a result of enhanced packing density.

$$\mu = \mu_0 + \frac{\eta\mu_0(\mu_m - \mu_0)}{N(1 - \eta)(\mu_m - \mu_0) + \mu_0} \quad (1)$$

where η : Volume packing fraction (= packing density), N : Coefficient of diamagnetic field, $0 \leq N \leq 1$, μ_m : Bulk permeability of powder, $\mu_0 = 4\pi \times 10^{-7}$ H/m

The behavior of Q reveals the same tendency as μ . As the composite core with S-Fe content increases to 30 wt.%, Q reaches a maximum value of 47 at 1 MHz. Q is proportional to reactance (ωL) and inversely proportional to resistance (R_s) from the equation. Since ωL is proportional to μ , Q increases as the μ of the mixed core increases, as shown in Fig. 5.

Core loss measurements also are required to predominantly evaluate performance of core materials to be applied to power applications. Figure 6 shows the frequency dependence of core loss at a magnetic flux density of 20 mT. We conducted these experiments at frequencies between 100 kHz and 3 MHz in order to evaluate the core material performance for inductor application. The core-loss increases logarithmically with an increase in driving frequency. For the total core loss driven at 1 MHz, we obtained a value of 562.2 mW/cm³, which was considerably low at the bimodal core of 7:3, representing the highest Q value. It is noteworthy that the addition of S-Fe having lower resistivity reduces the total core-loss for the entire frequency range.

$$P_{\text{tot}} = P_h + P_e + P_r \cong P_h + P_e \cong W_h f + W_e f^2 \quad (2)$$

where P_h : hysteresis loss, P_e : eddy-current loss, P_r : residual loss, W_h : hysteresis loss constant, W_e : eddy current loss constant.

Core loss can be classified into hysteresis loss, eddy-current loss, and residual loss, as indicated in Eq. 2. Usually, the magnetic metal core has a very low residual loss, which can be ignored.¹⁷ Calculated eddy current values driven at 1 MHz of the prepared cores are shown in Fig. 6b. Based on these results, the behavior of total core-loss largely depends on eddy-current loss than hysteresis loss for frequencies above MHz because eddy current loss is proportional to the square of the operating frequency. It is noteworthy that an increase in portion of the metallic S-Fe powders results to a increase in total core loss due to lower resistivity of

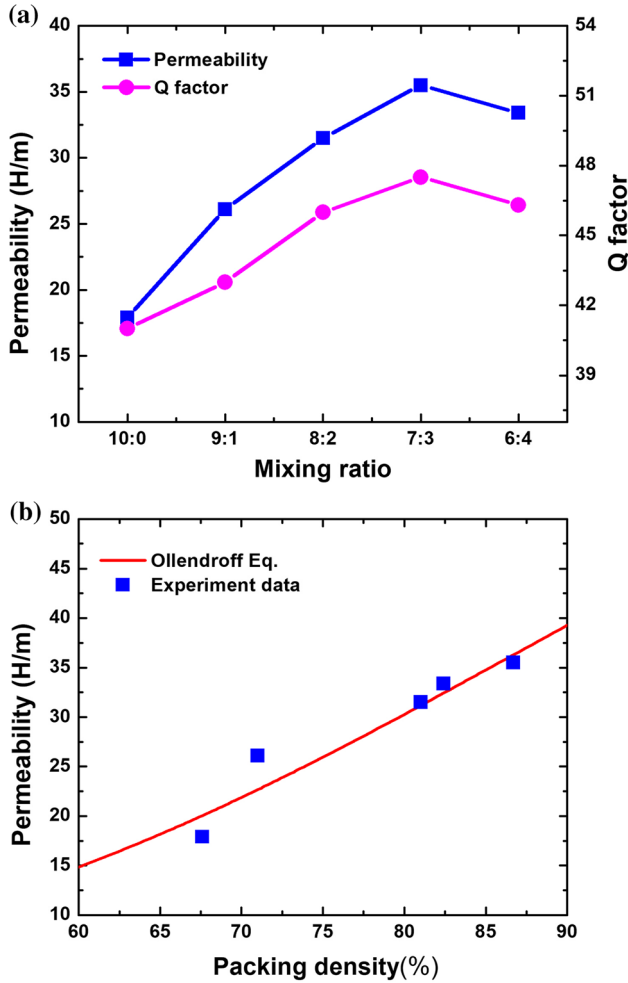


Fig. 5. The μ , Q (a) at 1 MHz of toroidal cores according to the mixing ratio and analysis of relation of permeability to packing density (b).

S-Fe powder resulting to a higher intra-particle eddy current loss. As the relation of packing density to Q , we can assume that this abnormal eddy current loss behaviors can also be induced from the optimized microstructure of the core. The result of low core loss can be attributed to an increase in Q induced as core packing density increase.

The permeability and core loss studied in this work and the previous results are presented in Table I. The maximum permeability of the core with the mixing ratio of 7:3 is higher than that of the core composed of FeCrSiB amorphous and carbonyl iron powder (CIP).^{5,18} Li et al.¹⁹ published experimental results of a bimodal core that was composed of FeSiB amorphous and permalloy (FeNi50). In that paper, the core mixed 85 wt.%FeSiB and 15 wt.%FeNi50 showed a 35 bellow magnetic resonance frequency range. The above comparison results order to compare core loss performance. For practical comparisons, the specifications of the commercial products (Mega flux, Fe-Si crystalline based alloy) currently applied to power inductors

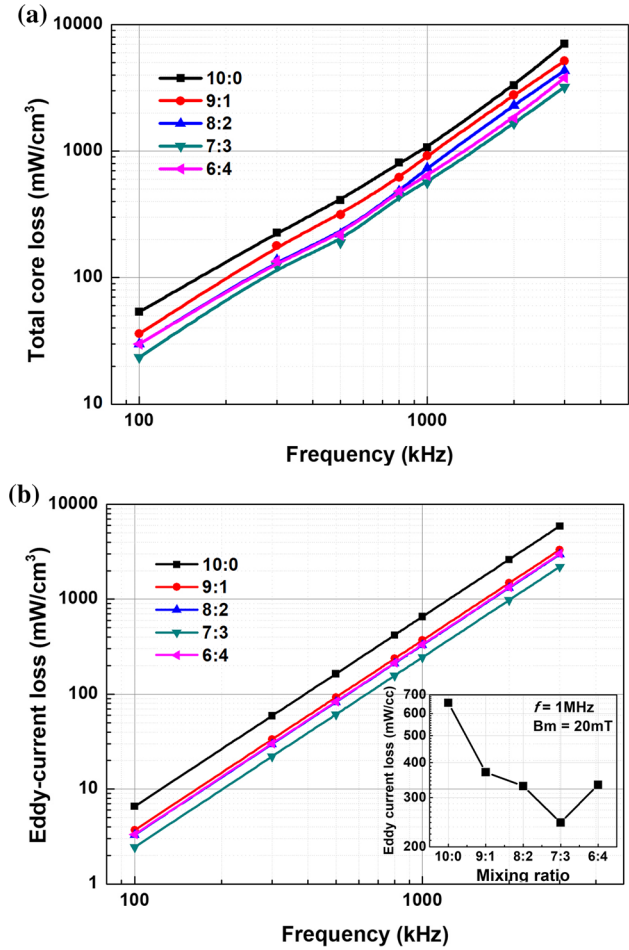


Fig. 6. Total core loss (a) and estimated eddy-current loss (b) of toroidal cores as a function of operating magnetic flux, 20 mT, in 100 kHz to 1 MHz. The inset shows the dependency of eddy current loss according to mixing ratio of powders (A-Fe alloy and S-Fe).

are presented in Table I. The cores with the mixing ratio of 7:3 in this study have half of core-loss of Megaflux at 100 kHz and 300 kHz. The comparisons demonstrate that the permeability of the core in this work has better permeability and core-loss than that of the other candidate, and will be applied to power applications working in the MHz.

CONCLUSION

In this study, we investigated the AC magnetic properties of soft magnetic composites with different mixing ratios of A-Fe and S-Fe powders in the MHz range. The core with a mixing ratio of 7:3 revealed a maximum packing ratio of 86.7%. The core with a maximum packing ratio showed a maximum permeability of 35.5, and Q of 47 evaluated at 1 MHz. Additionally, the core-loss was lowest at 562.2 mW/cm³, which was lower than commercial products applied to a power application as core materials. The analysis for the core's permeability calculated by Ollendorff's equation show that the permeability is totally dependent on the core packing density. These experimental results show that the core's

characteristics can be controlled not only by the intrinsic properties of the soft magnetic composites composing the core, but also by the microstructure, represented by the packing density. They also indicate that the maximum packing density of the core should be obtained for being the maximum permeability and minimum core-loss.

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